Functional Forecasting of Turkish Electricity Load Curves - Preliminary Report

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Purpose of the project

This project studies day-ahead forecasting of Turkish electricity demand. The raw quantity of interest is not a single daily number, but the full 24-hour load profile: how demand evolves overnight, rises in the morning, peaks during the day, and falls again in the evening. Treating each day as a curve preserves this intraday shape.

The practical question is whether a functional time-series model can improve on the available official day-ahead forecast. The report therefore compares each proposed model with two reference points: a simple weekly benchmark, Y_{d-7}, and the official forecast, $\widehat{Y}_d^{off}$. The weekly benchmark checks whether the model beats a very transparent rule based on the strong weekly rhythm of electricity use. The official forecast is the main benchmark because it represents the forecast that would be available in practice.

Object and functional representation

Let $Y_d(h)$ denote Turkish electricity load on day d, hour $h = 0, \ldots, 23$. The empirical object is the daily load curve

$$Y_d = \{Y_d(h) : h = 0, \ldots, 23\}.$$

The cleaned data are stored as daily curve matrices

$$Y = [Y_d(h)]_{d,h}, \qquad \widehat{Y}^{off} = [\widehat{Y}_d^{off}(h)]_{d,h}, \qquad E = [Y_d(h) - \widehat{Y}_d^{off}(h)]_{d,h},$$

with 24 hourly observations per retained day.

Each curve is represented by a B-spline basis,

$$Y_d(h) \simeq \sum_{m=1}^{M} c_{dm} B_m(h), \qquad M = 12.$$

FPCA gives

$$Y_d(h) = \mu(h) + \sum_{k=1}^{K} \xi_{dk} \phi_k(h) + e_d(h), \qquad K = 4.$$

Let

$$\xi_d = (\xi_{d1}, \ldots, \xi_{dK})', \qquad \Phi(h) = (\phi_1(h), \ldots, \phi_K(h))'.$$

Then

$$Y_d(h) = \mu(h) + \Phi(h)'\xi_d + e_d(h), \qquad \widehat{Y}_d(h) = \widehat{\mu}(h) + \widehat{\Phi}(h)'\widehat{\xi}_d.$$

All forecasts are rolling-origin forecasts: for target day d, estimation uses only information dated $< d$. The evaluation period starts on 2023-01-01 and contains 26,304 hourly observations.

Why these models

The model sequence is intentionally incremental. Each specification asks whether one additional source of information explains a remaining part of the forecast error.

First, the daily curve is reduced to a small number of FPCA scores. This is useful because the 24 hourly observations are highly dependent: morning, afternoon, and evening demand move together in recurring ways. FPCA summarizes the main level and shape movements of the whole curve, so the forecasting problem becomes forecasting a few daily scores rather than 24 separate hourly loads.

The first FPCA models use lagged scores. A weekly lag is natural because electricity load has a strong day-of-week structure: Mondays tend to resemble previous Mondays, weekends resemble previous weekends, and so on. Adding a one-day lag allows the model to react to recent changes in demand level or shape that are not fully captured by the same weekday one week earlier. This motivates the FPCA VAR(1,7) score model.

The next additions are calendar, holiday, and load-shape predictors. These variables are included because Turkish electricity demand changes around weekends, fixed holidays,religious holidays, bridge days, and seasonal school or summer periods. Lagged load-shape variables, such as the previous day's mean, peak, range, and morning ramp, are included because they describe the current demand regime before the target day is forecast.

Derivative features are then added to describe how quickly the load curve is changing within the day. Two days can have similar average demand but different ramping behavior. The derivative summaries and targeted morning, evening, and end-of-day ramps are meant to capture these slope and transition patterns.

Finally, the adaptive correction step addresses systematic residual patterns that remain after the score model. If recent forecasts have been too high or too low at particular hours, the correction uses those recent residual curves to adjust the base forecast. This is not a replacement for the FPCA model; it is a local post-processing step designed to correct short-run bias and intraday error patterns.

Forecasting specifications

Benchmarks:

$$\widehat{Y}_d^{SN}(h) = Y_{d-7}(h), \qquad \widehat{Y}_d^{OFF}(h).$$

Separate weekly score autoregression:

$$\xi_{dk} = \alpha_k + \beta_k \xi_{d-7,k} + u_{dk}, \qquad k = 1, \ldots, K.$$

Weekly FPCA VAR:

$$\xi_d = a + A_7 \xi_{d-7} + u_d, \qquad A_7 \in \mathcal{R}^{K \times K}.$$

FPCA VAR(1,7):

$$\xi_d = a + A_1 \xi_{d-1} + A_7 \xi_{d-7} + u_d, \qquad A_1, A_7 \in \mathcal{R}^{K \times K}.$$

Augmented score model:

$$\xi_d = a + A_1 \xi_{d-1} + A_7 \xi_{d-7} + B z_d + u_d,$$

where z_d is predetermined at date d.
Final corrected forecast:

$$\widehat{Y}_d^{final}(h) = \widehat{\mu}(h) + \widehat{\Phi}(h)' \widehat{\xi}_d + \lambda_d \widehat{r}_d(h).$$

Predictor construction

The largest predictor vector is

$$z_d = \left(z_d^{cal'}, z_d^{hol'}, z_d^{shape'}, z_d^{deriv'} \right)'.$$

Ramp and early-load variables:

$$R^m_{d-1} = Y_{d-1}(9) - Y_{d-1}(6), \qquad R^m_{d-7} = Y_{d-7}(9) - Y_{d-7}(6),$$

$$L^e_{d-1} = \frac{1}{3} \sum_{h \in \{5,6,7\}} Y_{d-1}(h).$$

Calendar controls:

$$z_d^{cal} = \left(weekend_d, \sin \frac{2\pi doy_d}{365.25}, \cos \frac{2\pi doy_d}{365.25} \right)'.$$

Holiday controls include fixed holidays, religious holidays, holiday eve, post-holiday, near-holiday, and bridge-day indicators:

$$Bridge_d = 1\{dow_d \in \{Mon, Fri\}\} 1\{d \notin \mathcal{H}\} 1\{d-1 \in \mathcal{H} \text{ or } d+1 \in \mathcal{H}\}.$$

Lagged load-shape variables:

$$\bar{Y}_{d-1} = \frac{1}{24} \sum_{h=0}^{23} Y_{d-1}(h), \qquad P_{d-1} = \max_h Y_{d-1}(h), \qquad M_{d-1} = \min_h Y_{d-1}(h),$$

$$G_{d-1} = P_{d-1} - M_{d-1}, \qquad \bar{Y}_d^{(q)} = \frac{1}{q} \sum_{j=1}^{q} \left(\frac{1}{24} \sum_{h=0}^{23} Y_{d-j}(h) \right), \quad q \in \{7, 28\}.$$

Derivative features are computed from the smoothed curve,

$$DY_d(h) = \frac{d}{dh} Y_d(h),$$

and summarized by

$$\overline{DY}_{d-1}, \quad SD(DY_{d-1}), \quad \max_h DY_{d-1}(h), \quad \min_h DY_{d-1}(h), \quad \max_h DY_{d-1}(h) - \min_h DY_{d-1}(h).$$

Targeted intraday changes:

$$R^{e}_{d-1} = Y_{d-1}(7) - Y_{d-1}(5), \qquad R^{ev}_{d-1} = Y_{d-1}(20) - Y_{d-1}(17), \qquad R^{end}_{d-1} = Y_{d-1}(23) - Y_{d-1}(20).$$

Adaptive correction uses past residuals

$$\widehat{e}_s(h) = Y_s(h) - \widehat{Y}^{base}_s(h), \qquad s < d,$$

and applies

$$\widehat{Y}^{corr}_d(h) = \widehat{Y}^{base}_d(h) + \lambda_d \widehat{r}_d(h),$$

where $\widehat{r}_d(h)$ is estimated from recent residual patterns and λ_d is selected on a validation window.

Evaluation

For model m,

$$e^{(m)}_{dh} = Y_d(h) - \widehat{Y}^{(m)}_d(h).$$

$$MAE_m = \frac{1}{N}\sum_{d,h} |e^{(m)}_{dh}|, \qquad RMSE_m = \left[\frac{1}{N}\sum_{d,h} (e^{(m)}_{dh})^2\right]^{1/2}, \qquad Bias_m = \frac{1}{N}\sum_{d,h} e^{(m)}_{dh}.$$

$$Gain^{MAE}_m = 100\frac{MAE_{OFF} - MAE_m}{MAE_{OFF}}, \qquad Gain^{RMSE}_m = 100\frac{RMSE_{OFF} - RMSE_m}{RMSE_{OFF}}.$$

Current results

The rows below should be read as a model-building path. The first two rows are benchmarks. Each later row adds one modeling idea: FPCA score dynamics, calendar and ramp information, derivative information, and then adaptive correction of recent residual errors.

Model	MAE	RMSE	Bias	MAE gain	RMSE gain
Seasonal naive Y_{d-7}	1912.46	3129.98	39.85	-57.60%	-79.85%
Official forecast	1213.50	1740.30	466.87	0.00%	0.00%
FPCA VAR(1,7) scores	1071.13	1562.67	-55.00	11.73%	10.21%
FPCA VAR(1,7) + ramp/calendar	1031.79	1493.24	112.34	14.97%	14.20%
FPCA VAR(1,7) + derivative features	890.01	1286.78	91.28	26.66%	26.06%
FPCA VAR(1,7) + holiday/load-shape + adaptive correction	666.49	996.22	66.46	45.08%	42.76%
FPCA VAR(1,7) + holiday/load-shape/derivative + adaptive correction	**640.10**	**967.82**	**103.39**	**47.25%**	**44.39%**

The final model improves on the official forecast by 47.25% in MAE and 44.39% in RMSE. Relative to the weekly seasonal naive benchmark,

$$100\frac{1912.46 - 640.10}{1912.46} = 66.53\%.$$